*Read me first*

# FORGE ONE

What it is, the risks, installing, and getting back



**FORGE ONE** · Felucca 1.6.1 + FORGE 0.13 · M-VAVE FM-1
github.com/kkasemsa/FORGE-ONE (forge branch) · 23 October 2026
Felucca by Hügelton Instruments and contributors · GPL-3.0-only




FORGE ONE is custom firmware for the **M-VAVE FM-1**. It is Felucca 1.6.1 by Hügelton Instruments, with one addition: **FORGE**, a seeded generator of melodies and beats on the SEQ TOOLS **F5** key, and its song starter, **STARTER**, on **G5**, and a live **PERFORM** mode on the FX layer. Everything else works exactly as in Felucca 1.6.1. FORGE ONE is an independent fork, not made or endorsed by Hügelton Instruments; it was called Felucca-FORGE until 0.5.1.

| | |
|---|---|
| Based on | Felucca 1.6.1 (GPL-3.0-only) by Leo Kuroshita, Hügelton Instruments |
| Adds | FORGE 0.13: SEED, DENSITY, EVOLVE and 8 STYLES on SEQ TOOLS F5, melodies and beats; STARTER on G5, 45 song starters in 13 genres with the 41-sound FORGE PACK, and SONG (F#5) for four parts A–D and the arrangement; PERFORM (FX double tap) for live mutes, solos, scenes and throws; the FORGE palette (changelog) |
| Branch | `forge` (`main` is an untouched copy of upstream) |
| Status | Builds and host tests pass in CI on every push. Installed and running on the author's FM-1 since 0.6; PERFORM (0.7) and the 45 starters (0.8) are not yet played on hardware. |

## Before you install

- **Installing firmware is at your own risk.** If an install is interrupted, the FM-1 can be left unable to start. You can bring it back (see If the FM-1 does not start), but that may need extra hardware.
- **Back up first.** The Felucca web editor can make a full backup of projects, presets and settings. The backup leaves out the FM6 voice bank, so keep your `.syx` file.
- **Your projects stay compatible.** FORGE only writes step data that Felucca 1.6 already stores, so projects made with FORGE open on stock Felucca 1.6 and 1.6.1.
- New to Felucca? Install stock Felucca first with the official web installer and get to know it, or try it in your browser.

## Get the firmware

1. Open the Releases page and download `forge-one.fwsc` from the newest release.
2. If there is no release yet, open Actions, click the newest green **Build** run on `forge`, and download the `forge-one-…` artifact. It is a zip, and `forge-one.fwsc` is inside. You need to be signed in to GitHub to download artifacts.

## Install (macOS)

The official web installer only installs stock Felucca, so install FORGE with Felucca's own command-line installer.

1. Quit Ableton Live and every other app that uses MIDI. Connect the FM-1 with a USB **data** cable and turn it on.
2. In Terminal, make a Python environment with the two MIDI libraries the installer needs. Do this once:
   - `python3 -m venv ~/fm1-env`
   - `~/fm1-env/bin/pip install mido python-rtmidi`
3. Download the installer script `tools/fm1_install.py` into the folder that holds `forge-one.fwsc`.
4. Run it from that folder: `~/fm1-env/bin/python fm1_install.py forge-one.fwsc` It asks before writing. Don't unplug the cable or let the Mac sleep until it says the install is done. The FM-1 restarts on its own.
5. Check it worked: the start-up screen says **FORGE ONE**, and when you hold **SEQ** the SEQ TOOLS map shows **FORGE** on key F5 and **STARTER** on G5.

## Build it yourself (optional)

GitHub builds every push, so you only need this to build your own changes. On a Mac with Docker Desktop running:

```
git clone --recurse-submodules -b forge https://github.com/kkasemsa/FORGE-ONE.git ~/FORGE-ONE
cd ~/FORGE-ONE
tools/forge_build.sh            # set up and build build/forge-one.fwsc
tools/forge_build.sh --flash    # build and install on the FM-1
```

The first run downloads JieLi's toolchain and three AC79 SDK files and makes a Python environment in `~/fm1-env`; later runs only build. `--test` also runs the host tests (about 10 minutes). See BUILDING.md for the details.

## Go back to stock

- **To stock Felucca:** use the official web installer in Chrome or Edge.
- **To the M-VAVE firmware:** use the web installer's **Return to official V15**, or M-VAVE's own updater.

## If the FM-1 does not start

- **If the install stopped part-way**, run the same install command again: it finishes the write.
- **If the screen stays black**, check whether your Mac sees a USB device named **WL80UBOOT**. That is the chip's own boot mode, and the FM-1 can be recovered. The command-line installer cannot reach it:

1. Try another USB data cable, quit every MIDI app, and run the official web installer again.
2. If it stays in boot mode, FM-1 Transporter can write the flash through a Seeed XIAO RP2040 wired to the FM-1's USB lines. Back up the flash first, then write the official firmware.

## The documents

| Document | What it covers |
|---|---|
| **Read Me First** (this page) | What this is, the risks, getting it on your FM-1, getting back |
| FORGE User Guide (PDF) | FORGE only: controls, the 8 styles, EVOLVE, recipes |
| FORGE ONE User Guide (PDF) | The whole instrument: panel, pages, sequencer, engines, effects, MIDI, menu, and FORGE |
| Tutorial (ภาษาไทย) (PDF) | A first 4-track jam: FORGE on tracks 1–3, drums by hand on track 4 |
| Felucca README | Upstream's own README, unchanged |
| BUILDING.md | Building the firmware yourself (CI does it on every push) |

## Credits and licence

FORGE ONE is by Khantipol (@kkasemsa): FORGE, STARTER, PERFORM and this fork's documents. It is built on Felucca:

Felucca is by Leo Kuroshita (@kurogedelic), Hügelton Instruments, and its contributors: see CREDITS.md. Its full manual is the Felucca wiki. On 16 October 2026 the maintainer stepped back from active development; Felucca stays available with occasional maintenance.

FORGE and this fork are free software under GPL-3.0-only, like Felucca. The bundled fonts, ported DSP and SDK files keep their own licences (LICENSES/, LICENSING.md). M-VAVE and FM-1 are trademarks of their respective owners. Neither Felucca nor this fork is affiliated with or endorsed by them.